EX.NO : 07
DATE : 20/08/2026

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL.

AIM :-

To write a program to implement flow control at the data link layer using the sliding window Protocol.

ALGORITHM :-

1) Start the program.
2) Read the total no. of frames sent (N), the senders window size (W) & a frame number.
3) Initialise base = 0 & nextstep = 0.
4) Repeat while base < N.
5) Repeat step-4 until all N frames have been transmitted & acknowledged.
6) Stop the program.

PROGRAM :-

```
def simulate (total_frames, window_size, last_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min (base + window_size,
                          total_frames - 1)
        print ("In. . . .")
```

```
Print (f"sender window: (Frame {base} - frame
                {window_end})")

while next_seq < base + window_size and
while next_seq < base + window_size and next_seq <
    total_frames :

    print (f" Node A → sending Frame {next_seq}")

if next.seq == lost_frame :

    print (f" (Frame {next_seq} lost in transit!)")

    next_seq += 1

if lost_frame is Not None and base <= lost_frame <
        next_seq :

    print (f"Node B --> Frame {lost_frame}
Missing Discarding out-of-order frames after it.)
Print (f"Node A --> sending Frame {next_seq}")
if next_seq == lost_frame :
    Print (f" (Frame {next_seq} lost in transit!)")
next_seq += 1
if lost_frame is not None and base <= lost frame <
                    next_seq :

    Print (f"Node B --> Frame {lost_frame} missing Discarding
out of order frames after it.")

    Print (f" Node A → Timeout ! Resending window starting from
    Frame {lost_frame}").

    next_seq = lost_frame

    lost_frame = None

    continue
```

```
for f in range(base, next_seq):
    print(f"Node B -> Recieved Frame {f}, sending Ack {f}")
    print(f"Node A -> Ack {next_seq-1} received sliding window forward.")
    base = next_seq
    print("\n....")
print(f"All {total_frames} frames successfully transmitted from Node A to Node B")
```

~~print~~

```
def main():
    print("... SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (GO BACK-N) ==\n")
    total_frames = int(input("Enter total no. of frames."),
    window_size = int(input("Enter sender's window size:"))
```

lost_frame = int(input(f"Enter frame number to simulate as lost in (0 to {total_frames-1}), enter -1 for no loss).

```
    if lost_frame == -1:
        lost_frame = None
    simulate(total_frames, window_size, lost_frame)
if __name__ == "__main__":
    main()
```

OUTPUT :-

Enter total no. of frames to send : 8

Enter sender's window size : 4

Enter frame number to simulate as LOST 10 to 7,
enter -1 for no loss : 5

Sender window : [Frame 0 - Frame 3]

Node A → sending Frame 0
Node A → Sending Frame 1
Node A → Sending Frame 2
Node A → Sending Frame 3
Node B → Received Frame 0, sending Ack 0
Node B → Received Frame 1, Sending Ack 1
Node B → Received Frame 2, sending Ack 2
Node B → Received Frame 3, sending Ack 3
Node A → Ack received, sliding window forward.

Sender window : [Frame 4 - Frame 7]

Node A → Sending Frame 4
Node A → Sending Frame 5

[Frame 5 lost in transit]

Node A → Sending Frame 6
Node A → sending Frame 7

Node B → Frame 5 missing. Discarding out-of-order frame after it.

Node A → Timeout! Resending window starting from Frame 5.

Sender window : [Frame 4 - Frame 7]

Node A → Sending Frame 5

Node A → Sending Frame 6

Node A → Sending Frame 7

Node B → Received Frame 4, sending ACK 4.

Node B → Received Frame 5, sending Ack 5.

Node B → Received Frame 7, Sending Ack 7.

Node A → Ack 7 received, sliding window forward.

All 8 frames successfully transmitted from Node A to Node B.

OUTPUT :-

Enter window size: 3
Enter number of frames to transmit: 5
Enter 5 frames: 1 2 3 4 5
With sliding window protocol the frames will be sent in the following manner (assuming no corruption of frames)

After sending 3 frames at each stage sender waits for acknowledgement sent by the receiver

1 2 3
Acknowledgement of above frames sent is received by sender

4 5
Acknowledgement of above frames sent is received by sender

RESULT :-

The program to implement flow control at the Data link layer using the sliding window protocol was successfully written & executed in python and it was verified by simulating the flow of frames b/w two nodes.